MATÍAS FERNÁNDEZ LAKATOS

Senior Neuromorphic Researcher · Machine Learning Engineer · Data Scientist

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PROFESSIONAL SUMMARY

Physicist with postgraduate training (Ph.D. and MSc in Physics, MSc in Big Data Analytics), currently **Senior Neuromorphic Researcher at Gradiant**, leading applied research on third-generation neural networks (Spiking Neural Networks), event-based vision, and energy-efficient AI at the edge. Previously developed intelligent systems for early anomaly and threat detection in corporate cybersecurity, with expertise in **unsupervised machine learning**, **real-time large-scale data processing**, and **explainable AI (XAI)**. More than 10 years of experience in research and teaching, with a strong background in statistical modeling, computer vision, and data-driven decision making. Neo4j Certified Professional with hands-on experience in **graph databases** and **graph data science**.

PROFESSIONAL EXPERIENCE

Senior Neuromorphic Researcher

Gradiant · Vigo, Spain · *Jul 2026 – present*

- Leading applied research on **third-generation artificial neural networks (Spiking Neural Networks, SNNs)**: new mathematical and computational models that mimic biological neurons and **synaptic plasticity**.
- Investigating SNN architectures for the analysis of different data types (**images, time series**), targeting the massive parallelism and **energy efficiency** required by AI-on-the-edge use cases.
- Researching frameworks and technologies for **compiling SNN models onto FPGAs**, as well as the design of specific neuromorphic hardware.
- Exploring how Gradiant's **photonic architectures** can evolve and be linked to neuromorphic computing.
- Investigating novel **explainable AI (xAI)** techniques applied to SNN-based models.
- Researching **neuromorphic vision sensors (DVS)** and how their event-based, low-throughput characteristics can be integrated into applications such as autonomous vehicles, manufacturing, and healthcare.
- Reviewing, expanding, and updating the **state of the art** in neuromorphic computing: new publications, hardware devices, and software tools, contextualized against project objectives.

Research Engineer

Gradiant · Vigo, Spain · *2025 – Jun 2026*

- Developed intelligent solutions for **early threat detection in corporate cybersecurity** using unsupervised ML models.
- Designed and implemented **β-Variational Autoencoder, Autoencoder, Extended Isolation Forest, and Isolation Forest** models for real-time anomaly detection and behavioral analysis on large-scale data environments.
- Focused on **model explainability (XAI)** to enhance usability for non-technical users and improve trust in AI systems.
- Integrated models into a scalable data architecture using **Kafka, Spark, and Airflow** for efficient real-time data stream processing.
- Tracked experiments and versioned models with **MLFlow**; managed large model artifacts with **MinIO**.
- Authored technical documentation for stakeholders (INCIBE) and collaborated with engineering/security teams in model integration and peer code reviews.

Researcher, Lecturer & Teaching Assistant

Faculty of Engineering, Universidad de la República (UdelaR) · Montevideo, Uruguay · *2015 – 2024*

- Taught **university-level physics** at the Physics Institute; developed and delivered course materials for undergraduate students.
- Conducted research in multiple projects: study of long-range properties of the strong nuclear interaction (QCD), development of multispectral instruments for remote monitoring, phase object retrieval, and creation of novel phase recovery and integration methods.
- Published **4 first-author peer-reviewed articles** in Scopus-indexed journals and 2 co-authored articles.
- Presented research at **international conferences** (SPIE, IOP, Elsevier), strengthening technical communication and presentation skills.

EDUCATION

MSc in Big Data Analytics Technologies

Universidad de Santiago de Compostela (USC) · Santiago de Compostela, Spain · *2024 – 2025*
Thesis: *Application of new anomaly detection models in Big Data contexts.* Coursework: Large-Scale Databases, Technologies for Unstructured Information Management, Computing Technologies for Big Data, IoT, Statistics, Data Mining, Data Visualization, Business Intelligence, Business Applications and Use Cases.

Ph.D. in Optics (Physics)

Universidad de la República (UdelaR) · Montevideo, Uruguay · *2019 – 2024* Thesis: Visualization and Characterization of Phase Objects Coursework: Coherent Optics, **Reinforcement Learning**, **Computer Image Processing**, **Computational Multivariate Statistics**, **Deep Learning for Computer Vision (CS231n)**, Fundamental Electronics Lab. Advisors: PhD. José A. Ferrari, PhD. Erna Frins, PhD. Gastón A. Ayubi.

MSc in Quantum Chromodynamics (Theoretical Physics)

Universidad de la República (UdelaR) · Montevideo, Uruguay · *2016 – 2018* Thesis: Role of the various couplings in infrared Quantum Chromodynamics Coursework: Statistical Mechanics, Quantum Field Theory I & II, General Relativity. Advisors: PhD. Nicolás Wschebor, PhD. Marcela Peláez.

Bachelor's Degree in Physics

Universidad de la República (UdelaR) · Montevideo, Uruguay · *2011 – 2015*

CERTIFICATIONS

Certification	Issuer	Date
Neo4j Certified Professional	Neo4j GraphAcademy	May 2026
Master Python for DevOps and CI/CD	Udemy	2025
Using Neo4j with Python	Neo4j GraphAcademy	May 2026
Intermediate Cypher Queries	Neo4j GraphAcademy	May 2026
Importing Data Fundamentals	Neo4j GraphAcademy	May 2026
Graph Data Modeling Fundamentals	Neo4j GraphAcademy	May 2026
Cypher Fundamentals	Neo4j GraphAcademy	Apr 2026
Neo4j Fundamentals	Neo4j GraphAcademy	Apr 2026

Full Neo4j profile: graphacademy.neo4j.com/u/34d98b08-78b3-4f35-a97c-3c869a7dec4e

TECHNICAL SKILLS

Neuromorphic Computing (current research)

Spiking Neural Networks (SNNs) · Synaptic Plasticity · Event-based Vision (DVS sensors) SNN-to-FPGA Compilation · Photonic Neuromorphic Architectures · Energy-efficient / Edge AI · xAI for SNNs

Machine Learning & AI

`scikit-learn` · `PyTorch` · `TensorFlow` · `Keras` · `MLFlow` Unsupervised Learning · Anomaly Detection · Autoencoders · Variational Autoencoders · Isolation Forest Explainable AI (XAI) · Reinforcement Learning · Deep Learning · Computer Vision

Data Engineering & Big Data

`Apache Kafka` · `Apache Spark` · `Apache Airflow` · `MinIO` Real-time Stream Processing · Large-Scale Data Pipelines · ETL · IoT Data

Graph Databases

`Neo4j` · `Cypher` · Graph Data Modeling · Graph Data Science Knowledge Graphs · Network Analysis

Programming & Tools

`Python` · `R` · `SQL` · `MATLAB` · `LaTeX` `Git` · `Linux` · `Docker` (basic)

Data Analysis & Visualization

`pandas` · `NumPy` · `SciPy` · `matplotlib` · `seaborn` · `plotly` Statistical Analysis · Multivariate Analysis · Optimization · Signal Processing

Computer Vision & Image Processing

`OpenCV` · `scikit-image` · `PIL/Pillow` Phase Object Visualization · Digital Image Processing · Pattern Recognition

PUBLICATIONS

First-Author Articles (Peer-Reviewed, Scopus Indexed)

1. **[2024 – SPIE]** Integration with one partial derivative applied to quantitative phase imaging
2. **[2023 – Optics and Lasers in Engineering, Scopus]** Common-path quantitative phase imaging by propagation through a sinusoidal intensity mask
3. **[2022 – Optik, Scopus]** Phase retrieval by amplifying the prism term of the Transport of Intensity Equation with a sliding step function
4. **[2019 – IJMPA, Scopus]** On the contribution of different coupling constants in the infrared regime of Yang–Mills theory

Co-Authored Articles

1. **[2025 – Optics & Laser Technology]** Wavefront reconstruction of discontinuous phase objects using directional derivatives
2. **[2024 – IOP Science / Measurement Science and Technology]** Hilbert's and Takeda's single-shot interferometry with a linear-carrier: a comparison

ACHIEVEMENTS & FELLOWSHIPS

- **Member of the National System of Researchers (SNI)**, Uruguay, 2024
- **PhD Scholarship**, Comisión Académica de Posgrados (CAP), 2022
- **PhD Scholarship**, Agencia Nacional de Investigación e Innovación (ANII), 2019
- **Master's Scholarship**, Comisión Académica de Posgrados (CAP), 2016

LANGUAGES

Language	Level
Spanish	Native
English	Professional working proficiency

PERSONAL PHILOSOPHY

"*Journey before destination*" — a mindset for continuous learning, creative exploration, and embracing the process as much as the outcome.

Last updated: July 2026